



Renewable vs. Nonrenewable

S6E6a

A **natural resource** = anything from the Earth that people use (water, air, soil, metals, fuels). The big sorting question: **can it refill itself faster than we use it?**

RENEWABLE — refills naturally

Won't run out if used wisely = **sustainable**:

Solar • **Wind** • **Hydro** (water) • **Geothermal** (Earth's inner heat) • **Biomass** (burning plant material) • **Tidal**

NONRENEWABLE — limited supply

Takes **millions of years** to form, so it **CAN** run out:

Coal • **Oil** • **Natural gas** (= the 3 **fossil fuels**) • **Nuclear** (uranium, a mined metal)

Fossil fuels formed from the buried remains of ancient plants and tiny sea creatures, squeezed and heated underground for millions of years. **The trade-off**: they're powerful and cheap for now — but they'll run out, AND burning them releases greenhouse gases that warm the planet.

ACTIVITY 1: R or N?

Write **R** (renewable) or **N** (nonrenewable):

- | | |
|---------------------------------------|--------------------------------------|
| _____ 1. Solar panels on a roof | _____ 5. Heat from inside the Earth |
| _____ 2. Coal burned at a power plant | _____ 6. Nuclear energy from uranium |
| _____ 3. Wind turbines on a hill | _____ 7. A dam using flowing water |
| _____ 4. Gasoline (made from oil) | _____ 8. Natural gas heating a home |

Conserving Water, Soil & Air

S6E6b

Conservation = using resources carefully so they last and stay clean for the future.

Resource	The Problem	Ways to Protect It
Water	Fresh water is limited; pollution makes it unsafe	Fix leaks, shorter showers, keep chemicals out of drains & rivers
Soil	Erosion & over-farming strip away fertile soil	Plant cover crops & trees, rotate crops, don't clear too much land
Air	Burning fuels releases pollution & greenhouse gases	Use cleaner energy, drive less, reduce factory emissions

The Three R's: **Reduce** (use less) • **Reuse** (use things again) • **Recycle** (turn old materials into new products). Each one pulls fewer new resources from the Earth!

ACTIVITY 2: Which R Is It?

Write **REDUCE**, **REUSE**, or **RECYCLE**:

- | | |
|--|---|
| 1. Refilling a water bottle instead of tossing it: _____ | 4. Turning off lights when you leave: _____ |
| 2. Taking shorter showers: _____ | 5. Donating clothes for someone else to wear: _____ |
| 3. Old cans melted into new metal products: _____ | 6. Paper collected and made into new paper: _____ |

Flip over for the warming planet — and what we can do! →



A Warming Planet

S6E6c

Over the past century, Earth's **average temperature has slowly but steadily risen**. The line wobbles year to year, but the long-term trend climbs. Here's the connection:

Burning fossil fuels → releases extra greenhouse gases (especially CO₂) → more gases trap MORE heat → average temperature rises



The Evidence

Scientists check many lines of evidence: **temperature records**, **melting glaciers**, **rising sea levels**, and the amount of **CO₂ in the air** — and the data all point the same way. The temperature rise lines up closely with greenhouse gases from human activity.



Remember from Ch. 4

Greenhouse gases work like a **blanket**: sunlight in, some heat trapped. Without any, Earth would freeze. The concern is **EXTRA** gases trapping **TOO MUCH** heat.



ACTIVITY 3: Connect the Chain

Fill in the blanks with: *CO₂*, *heat*, *fossil fuels*, *risen*, *glaciers*

1. Burning _____ releases greenhouse gases
2. The main greenhouse gas released: _____
3. Greenhouse gases trap _____ near Earth
4. Over the past century, average temperature has _____
5. Melting _____ are one piece of evidence
6. Another piece of evidence: rising _____ levels



What We Can Do

S6E6c

Good news: human choices are part of the cause, so human choices can be part of the **solution**. Big actions AND small habits add up:

BIG Solutions (countries & companies)

Switch to renewable energy (solar, wind, hydro)
Build more efficient vehicles & buildings
Protect & plant forests (trees absorb CO₂)
Reduce factory pollution

EVERYDAY Habits (you!)

Turn off lights & unplug devices
Walk, bike, carpool, or use the bus
Reduce, reuse, recycle
Save water & electricity at home



Trees are natural helpers: plants take in CO₂ and release oxygen — protecting forests and planting trees removes greenhouse gas from the air while making the oxygen we breathe!



ACTIVITY 4: Resource Round-Up

1. Energy from splitting uranium atoms: _____
2. "Geo" + "thermal" literally means: _____
3. The 3 fossil fuels: _____, _____, _____
4. Able to be used without being used up: _____
5. One way trees help the air: _____
6. Using resources carefully so they last: _____



I Can... (Self-Check)

- | | |
|--|--|
| ☐ Define "natural resource" and sort renewable vs. nonrenewable | ☐ Give ways to protect water, soil, and air |
| ☐ Name 6 renewable energy sources | ☐ Explain the Three R's with examples |
| ☐ Name the 3 fossil fuels and how they formed | ☐ Connect fossil fuels → CO ₂ → rising temperature |
| ☐ Explain why nuclear energy is nonrenewable | ☐ Name evidence of warming & solutions big and small |